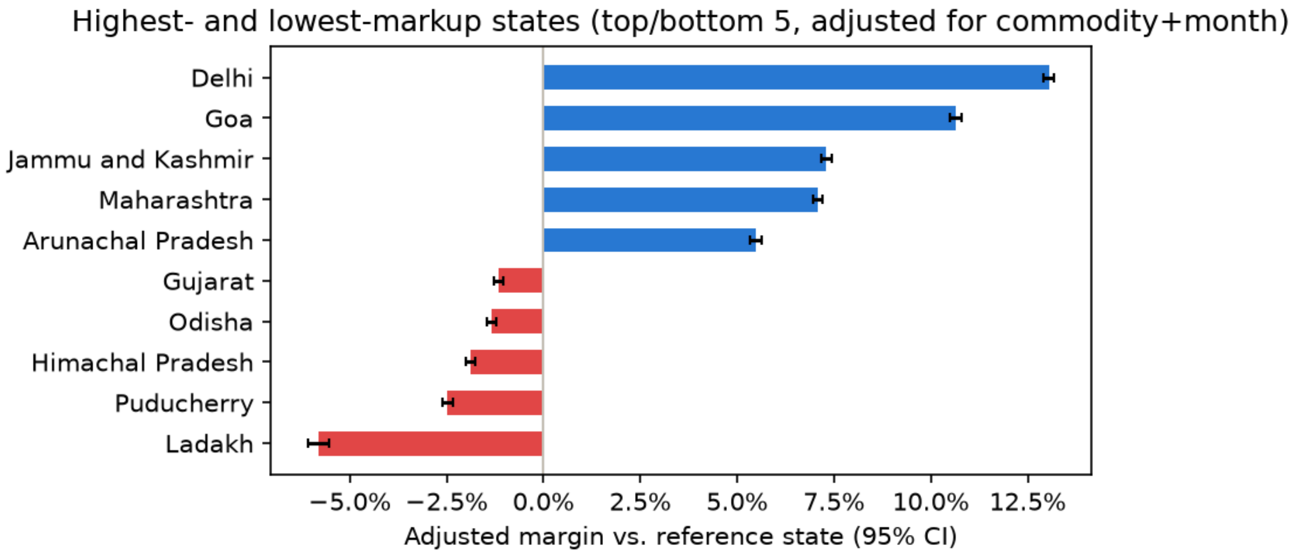
Retail Markups on Essential Commodities Vary Sharply by State

A statistical analysis of 2.05 million state-commodity-day observations, 2014-2026



The problem

When wholesale prices rise, retail prices don't always rise with them — and by how much they diverge differs by where you live. If the gap between what a shopkeeper pays a wholesaler and what a consumer pays at the till is systematically larger in some states than others, that is either a sign of local market inefficiency, or of consumers in those states quietly absorbing a larger middleman margin. Either way, it is a question regulators can act on — but only if the disparity is real and not noise.

What we did

We combined daily retail and wholesale prices for 17 essential commodities (onion, wheat, rice, sugar, milk, pulses, edible oils, tea, salt) across all 35 Indian states and union territories, 2014–2026, published by the Dept. of Consumer Affairs. For each state-commodity-day we computed the retail markup over wholesale price, then asked whether that markup differs by state — first with a distribution-free test per commodity (Kruskal-Wallis, since markups are right-skewed, not normally distributed), then with a regression that adjusts for which commodity and which month it is, so the comparison isn't just picking up commodity mix or seasonal effects. Every step — cleaning, the primary test, the regression, and two robustness checks — runs from a single seeded script; nothing here comes from a one-off notebook calculation.

What we found

The markup differs by state in every one of the 17 commodities tested, even after adjusting for which commodity and which month it is (Kruskal-Wallis per commodity, Benjamini-Hochberg corrected across all 17 tests — none survive by chance). The size of the effect varies by commodity, from smallest (wheat) to largest (iodised salt), but the direction is consistent: **which state you're in predicts your markup, on top of what you're buying and when.**

The regression puts a number on it. Holding commodity and month fixed, **Delhi's markup runs about 18.8 percentage points higher than Ladakh's** (95% CI for each state's estimate is within ±0.15 points — the gap itself is not a coincidence of a small sample). Delhi, Goa, Jammu & Kashmir, Maharashtra, and Arunachal Pradesh have the five highest adjusted markups; Ladakh, Puducherry, Himachal Pradesh, Odisha, and Gujarat have the five lowest. This regression explains 35.9% of the variation in markups overall (R^2=0.359) — state, commodity, and month together are real, substantial predictors, not the whole story.

In plain language: across all 2.05 million observations, the typical (median) Indian consumer pays a **9.3% markup** over the wholesale price for these commodities (95% bootstrap CI: 9.29%–9.31%). That's the baseline. State of residence alone shifts a consumer up or down that baseline by amounts large enough to matter for a household budget — nearly 19 percentage points between the highest- and lowest-markup states, for the same commodity in the same month.

(Exploratory, not part of the pre-registered test above:) onion prices are 4.5× more volatile day-to-day than other commodities — consistent with onion being the commodity most frequently subject to export bans and stock-limit interventions, though this analysis cannot say whether the volatility causes the intervention or the reverse.

Recommendation

State consumer-affairs departments in the five highest-markup states — starting with Delhi, where the adjusted gap is largest and the sample (n>4,000 days for onion alone) is not in doubt — should investigate retail supply-chain structure for the commodities driving their state's premium, rather than treating "retail prices are higher here" as an unavoidable fact of local logistics. This dataset identifies *where* to look; it does not by itself identify *why*, which is exactly where an audit or a follow-up study should start.

Limitations

- **Correlational, not causal.** This analysis identifies *that* markups differ by state, not *why*. Transport cost, spoilage/perishability, local demand elasticity, and retail market concentration are all plausible drivers and none are in this dataset — state could be standing in for any of them.
- **State-level, not sub-state.** The underlying data reports one retail and one wholesale price per state per day, not per city or market — a state average can mask large within-state variation (a state capital vs. a rural district, for instance).
- **Real missingness, not imputed.** ~5.8% of state-commodity-days are missing a price in the underlying data (some states/commodities more than others — up to ~21% for edible oils in a few states); these are excluded from the relevant test, not filled in, so the smallest states/commodities carry less statistical weight by construction.
- **External validity.** This is an India-specific finding, using India-specific market structures and a India-specific state-boundary history (state definitions changed mid-panel: J&K split into J&K + Ladakh in 2019, Dadra & Nagar Haveli merged with Daman & Diu in 2020) — it should not be assumed to generalize to other countries' retail-wholesale systems.
- **The onion-volatility finding is exploratory**, pre-registered as secondary in the project's analysis plan specifically so it would not be presented with the same confidence as the primary state-effect finding above.

Every number in this brief is reproducible from `reports/results.json`, generated by `make analysis` against the pipeline in this repository. Full methodology, data provenance, and code: see `PROBLEM.md` and `data/README.md`.